Kritansh Uppal

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EDUCATION

Bachelor of Technology in Computer Science & Engineering *Aug 2024 – May 2028(Expected)*
Punjab Engineering College, Chandigarh **CGPA: 8.82/10**

EXPERIENCE

TBRL, DRDO - Research Intern *May 2026 - Jul 2026*

- Built a CV pipeline (**YOLO**, **ByteTrack**, optical flow, geometric methods) for fragment detection, tracking, and velocity estimation from high-speed imagery - achieving **70%** detection with velocity accuracy within **±10%** after calibration.
- Re-engineered the pipeline for **air-gapped, offline deployment** - bundling all 70 dependencies into a standalone Windows installer requiring zero internet connectivity.

Zimo One - Software Intern *Jun 2025 - Jul 2025*

- Resolved **15+ production bugs** in Node.js backend services, improving internal tool uptime by **12%** for workflows used by **30+ team members**.
- Implemented REST API validation using schema-based checks, reducing regression issues by **25%** across multiple services.

PROJECTS

SkillForge AI - Adaptive Onboarding Engine *Mar 2026*
FastAPI, React, NLP, LLMs, NetworkX, SentenceTransformers, PDF Parsing Source Code

- Resume-to-role matching is manual and subjective, so built **SkillForge AI** - a pipeline (parsing, skill extraction, gap analysis, path generation) that auto-generates personalized roadmaps.
- Combined **keyword + SentenceTransformer** skill extraction with a **weighted gap-analysis system**, then sequenced results via a **dependency-aware learning engine (NetworkX)**.
- Delivered explainable JSON outputs via **FastAPI + React**, replacing ad-hoc advice with a reproducible recommendation.

Automatic Short-Form Video Generator (Vizard-AI Replica) *Dec 2025 - Jan 2026*
Computer Vision, Multimodal Systems, Python Source Code

- Long-form-to-shorts editing is manual and slow, so architected an automated clipping pipeline, cutting editing time by**70%**.
- Compared uniform vs. scene-change-based frame sampling and selected adaptive key-frame extraction, improving clip relevance by **22%**.
- Validated the pipeline at scale by processing **5+ hours of video per run** using reproducible, CV-based frame extraction.

ProfitLens - Amex Campus Challenge 2026(1st at PEC, Overall 7th -> Round 1) *Jun 2026 - Jul 2026*
Streamlit, Gemini, RAG, FAISS, Pandas Source Code

- Needed to explore the 500K-row scored Amex Premier dataset without risking a chatbot inventing numbers, so built **ProfitLens**: a Streamlit + Gemini assistant combining function-calling tools (live pandas reads) with a local **RAG** layer (FAISS + Gemini embeddings) grounded on the feature dictionary.
- Designed the scoring pipeline around a single frozen weights dict, so changing one coefficient regenerates both predictions and the human-readable formula together – iterated to a **93.44%** public-leaderboard score in Round 1.
- Kept every answer accurate and cited rather than invented, streamlining iteration across the team's submission waves.

Intelligent Data Extraction from Business Reports (1st / 50+ Teams, PECathon 2025) *Nov 2025*
Computer Vision, Document AI, YOLO, OCR, RAG

- Manually reviewing unstructured business PDFs for tables, charts, and context is slow, so led development of a document-intelligence system combining layout analysis, OCR, and a **RAG** pipeline for summarization and Q&A.
- Achieved **90%+ table and chart detection accuracy** through layout segmentation and image preprocessing.
- Shipped an end-to-end automated document-analysis system that secured **1st place** among 50+ teams.

TECHNICAL SKILLS

Programming: Python, C++, SQL
ML / AI: Machine Learning, NLP, LLMs, RAG, Feature Engineering, Data Analytics
Frameworks & Libraries: FastAPI, Scikit-learn, LangChain, FAISS, Pandas, NumPy, Firebase, Microsoft Suite
Data & Product: MongoDB, BigQuery, Competitive Analysis, Data Pipelines, Risk Management, Data Governance
Tools & Systems: Git, Linux, REST APIs, Docker, Azure, AWS

ACHIEVEMENTS

- **Winner, PECathon 2025**- Open-to-all inter-college hackathon conducted during PECFEST with 50+ teams.
- CodeChef Rating: **1568**, Codeforces Rating: **1329**, LeetCode Contest Rating: **1583**.
- **Awarded by Punjab Governor Sh. Gulab Chand Kataria ji** for excellence in sports.
- **State Gold Medalist**- Table Tennis, Chandigarh.

LEADERSHIP EXPERIENCE

- **Creative Head, WOPI**- Led branding and visual strategy for an international academic conference.
- **Assistant Cultural Clubs Secretary (ACCS), PEC-** Liaise between student cultural clubs and college administration, coordinating official cultural initiatives.